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Repeated spinal flexion modulates the flexion–relaxation phenomenon

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Abstract

Objectives. To determine if repeated spinal flexion and loading modulate the deactivation of lumbar muscles near full flexion (flexion–relaxation).

Design. Repeated measures experimental study of the effect of repetitive trunk flexion and added mass on the flexion–relaxation phenomenon.

Background. Repeated flexion causes muscular fatigue, creep of passive tissues and diminished protective reflexes. However, flexion–relaxation has not been studied in repeated trunk flexion, and could be related to the increased risk of low-back disorders.

Methods. Thirty healthy young subjects performed 100 trunk flexion movements between standing and full flexion. Erector spinae electromyography and lumbar spine flexion were measured during cycles 1–10 (no load), 11–20 (performed holding a mass in the hands), 81–90 (mass in the hands) and 91–100 (no load). The spinal flexion angle at myoelectric silence and full flexion were extracted from each movement cycle.

Results. Twenty-three of the 30 subjects showed flexion–relaxation throughout the repeated trunk flexion. The flexion–relaxation and maximum flexion angles increased at the end of the experiment; the flexion–relaxation angle relative to the maximum flexion angle also increased. This effect depended on the load condition; the flexion–relaxation and maximum flexion angles showed a greater increase in the unloaded than loaded condition.

Conclusions. The flexion–relaxation phenomenon was changed due to repeated trunk flexion. The increases in flexion–relaxation angle likely involve changes to the neuromuscular control system.

Relevance

The deactivation of the erector muscles near full flexion occurs at a greater spinal flexion angle and a greater proportion of maximum spinal flexion following repeated spinal flexion. This may be related to the increased risk of injury associated with repeated flexion.
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Keywords: Electromyography; Erector spinae; Flexion–relaxation; Lumbar; Spine; Repeated loading

1. Introduction

Epidemiologic studies illustrate that repeating bending and lifting activities are associated with increased risk of developing low-back disorders (Marras et al., 1995; Xu et al., 1997). Trunk flexion generates large bending moments in the passive spinal tissues (Dolan et al., 1994a; Adams and Dolan, 1991) and large compressive loads due to muscular forces (Dolan et al., 1994a; Schultz et al., 1985). In vitro studies show that repetitive flexion with concurrent compression can cause prolapsed disc injuries (Callaghan and McGill, 2001). Repeated flexion causes muscular fatigue (Sparto et al., 1997, Dolan and Adams, 1998), creep of the intervertebral disc and ligaments (Adams and Dolan, 1996, Wilke et al., 1998) and diminished protective muscular reflexes (Solomonow et al., 1999). Repeated bending has been hypothesized to increase the risk of injury to the intervertebral disc as the viscoelastic changes in the ligaments are thought to occur more rapidly than the

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disc and accordingly the disc is thought to bear a greater proportion of the flexion moment (Dolan and Adams, 1998).

The erector spinae are derecruited as subjects approach full lumbar flexion (Floyd and Silver, 1951). This phenomenon is called flexion–relaxation, and is particularly noteworthy as the external moment is borne by the passive spinal tissues (Golding, 1952; Kippers and Parker, 1984; Andersson et al., 1996; McGill and Kippers, 1994). Many studies report flexion–relaxation in static trunk flexion postures (Floyd and Silver, 1951; Golding, 1952; Kippers and Parker, 1984; Andersson et al., 1996; McGill and Kippers, 1994; Sarti et al., 2001; Wolf et al., 1979; Floyd and Silver, 1955). Recent studies report that the flexion–relaxation phenomenon also occurs during lifting (Mathieu and Fortin, 2000; Toussaint et al., 1995; Dolan et al., 1994b), lateral bending (Raftopoulos et al., 1988) and during sitting (Callaghan and Dunk, 2002). The flexion–relaxation phenomenon is modulated by the amount of load lifted (Schultz et al., 1985; Kippers and Parker, 1984; Gupta, 2001), lifting rate (Sarti et al., 2001) and differences have been reported between low-back pain and healthy subjects (Golding, 1952; Sihvonen et al., 1991; Ahern et al., 1988; Shirado et al., 1995). Previous studies have not evaluated the effect of repeated flexion on the mechanics of the flexion–relaxation phenomenon, which may be a factor in the increased risk of injury with repeated flexion.

The purpose to the current study was to assess the effects of repetitive bending on the flexion–relaxation phenomenon in order to evaluate whether changes to the tissue (passive and muscle) mechanical properties and/or neuromuscular control system occur due to repetitive bending. Our hypotheses were that repetitive bending would result in increased peak flexion during the lift and parallel increases in the angle for onset of the flexion–relaxation phenomenon. Additionally we hypothesized that adding a mass will result in greater peak flexion during the lift and greater angle for onset of the flexion–relaxation phenomenon.

2. Methods

2.1. Participants

Thirty healthy and active young adults (15 males and 15 females) were recruited from a university student population (Table 1). Written consent to participate in the investigation was obtained from all subjects, and an Institutional Review Board approved the procedure for this project.

Table 1
Mean and (standard deviation) for the 30 subjects' anthropometrics

Subject	Age (years)	Height (m)	Mass (kg)
Female ($n = 15$)	20.4 (0.8)	1.65 (0.06)	61.4 (7.1)
Male ($n = 15$)	21.4 (1.4)	1.78 (0.07)	79.6 (11.8)

2.2. Instrumentation

Disposable Medi-Trace surface electromyography (EMG) electrodes (Ag–AgCl) were applied bilaterally to the skin 3 cm from the midline at the level of T9 (thoracic erector spinae; TES) and L4/L5 (lumbar erector spinae; LES). The signals were differentially amplified (Bortec AMT8, Calgary, AB, Canada), band-pass filtered (10–1000 Hz), and converted to digital using a 12-bit system (National Instruments PCI-6023E, Plano, TX, USA) at a rate of 1000 samples/s. The EMG data was full-wave recitified and low-pass filtered at 6 Hz and down-sampled to 20.5 Hz.

The lumbar flexion–extension angle was measured using a 3-Space ISOTRAK (Polhemus, Colchester, VT, USA) and the output was sampled at 20.5 Hz, synchronously with the EMG. The electromagnetic source was strapped over the sacrum and the sensor was attached to the skin overlying the T6 spinous process. As suggested by Potvin (1992), the resulting angles were multiplied by 0.77 to estimate the angle between L1 and the sacrum. Relaxed standing was used as the zero position for lumbar flexion.

2.3. Data collection

Initial prone resting and standing maximal voluntary contraction (MVC) trials were collected for the LES and TES muscles. In terms of the MVC trials, the subjects stood with their pelvis and upper back secured such that they could perform maximum trunk extension efforts without any resulting movement. The apparatus used for this purpose was very secure such that subjects would feel comfortable and safe performing maximal efforts. The subjects performed three maximal extension efforts, and three trials in which they also retracted their scapulae. During these trials, subjects attempted to elicit the maximum electrical activity from the upper and lower erector spinae muscles (LES and TES). Several minutes of rest were allowed between MVC trials. The largest voltage for each of the EMG channels across all of the MVC trials was selected as maximal for subsequent normalizations.

The subjects performed a series of 100 consecutive full trunk flexion and return to standing maneuvers at a fixed rate of 11 s per trunk flexion cycle. The cycle was divided into three phases, and was indicated using audible tones: flexion (4.5 s), hanging (2 s) and extension to standing (4.5 s). The sequence of one hundred trunk flexions included trials with and without additional load; the loads were held in the hands throughout the cycle. The loads were calculated using psychophysical load

limits (Snook and Ciriello, 1991); 10 kg for male and 5 kg for female subjects. The subjects performed ten repetitions with no-load, ten repetitions with the full load, sixty repetitions with half of the load (5 kg for male and 2.5 kg for female subjects), ten repetitions with the full load, and 10 repetitions with no-load. All subjects performed the same sequence. Data was collected during the initial and final 20 lifts (no load/loaded conditions).

2.4. Analysis

The initial resting amplitude was subtracted from each of the EMG channels, and the amplitudes were normalized to percent MVC. The EMG and spinal flexion patterns were displayed using custom software (Labview, National Instruments, Plano, TX, USA). The phases of electrical activity and silence were manually identified from each trace (Fig. 1). The following events were identified for each flexion–extension cycle: onset of muscle activation during lowering, onset of flexion–relaxation, maximal trunk flexion (the maximum lumbar flexion angle), onset of muscle activity during raising and the end of muscle activity during raising (Fig. 1). The current study focused on the EMG amplitude and the magnitude of spinal flexion at the onset of flexion–relaxation. The flexion–relaxation angle was also expressed as a percentage of the maximum flexion angle for each cycle. Averages of the 10 repeated trials were taken for each variable. A threshold of 1% activation was used to define flexion–relaxation; subjects who did not achieve this degree of muscular relaxation in the lumbar erector spinae were excluded. A three-way repeated measures ANOVA (time $*$ load $*$ gender) was used to determine experimental effects ($P < 0.05$ was considered significant). Post hoc tests were performed with paired t-tests using a Bonferroni correction such that the familywise error remained at $P < 0.05$.

3. Results

Flexion–relaxation was demonstrated throughout the experiment in 21 of the 30 subjects (10 males and 11 females). The remaining subjects showed marked reductions in lumbar muscle activation near full spinal flexion, though the reductions were not sufficient to meet our stringent criterion defining flexion–relaxation.

There were no significant effects of gender observed in the study. We observed significant changes in the flexion–relaxation response with repeated spinal flexion. For maximum flexion parameter there was a significant effect of load ($P < 0.01$) and an interaction between time and load ($P < 0.01$, Fig. 2). Post hoc tests indicated no effect of repetitive flexion when loaded, but a significant increase occurred when unloaded ($P < 0.05$). For the flexion–relaxation angle parameter, there was a significant interaction between load and time ($P < 0.05$), with repetitive flexion resulting in higher flexion–relaxation angles only in the unloaded condition (Fig. 2). Interestingly, for flexion–relaxation angle expressed as a percent of the maximum flexion angle, there was a main effect of time ($P < 0.05$, Fig. 3) but no significant effects or interactions of gender or load.

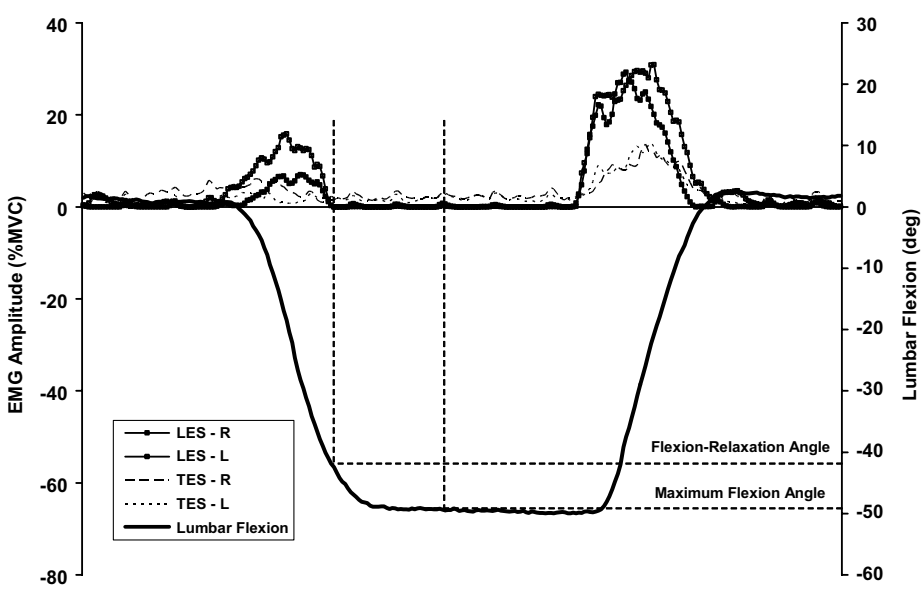

Fig. 1. Typical lumbar flexion and EMG patterns for one flexion–extension cycle. The EMG amplitudes have been normalized to maximal voluntary contraction, and are measured with respect to the left-hand scale. The lumbar flexion angle is measured with respect to standing and described using the right-hand scale. The following events are identified: onset of flexion–relaxation and maximal trunk flexion. This trial occurs over 13.3 s.

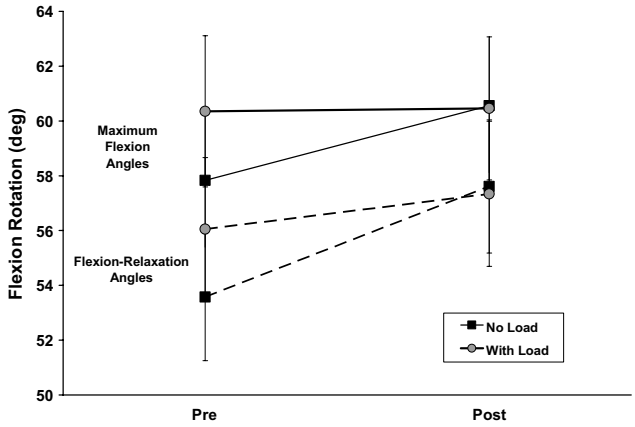

Fig. 2. Flexion–relaxation angle and maximum flexion angle as a function of time (pre/post) and load condition. There was a statistically significant interaction between load and time for both the flexion–relaxation and maximum flexion angles. There was no significant gender effect, so the data was pooled across genders (10 males and 12 females). The mean and standard error are shown ($n = 21$).

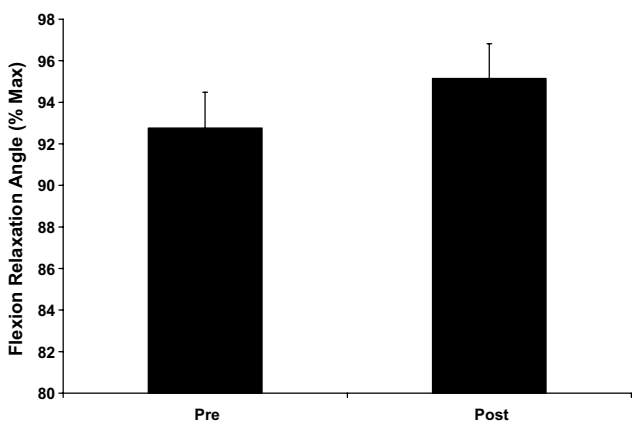

Fig. 3. Effect of time (repetition) on the flexion–relaxation angle stated as a percentage of maximum flexion. The mean and standard error are shown ($n = 42$).

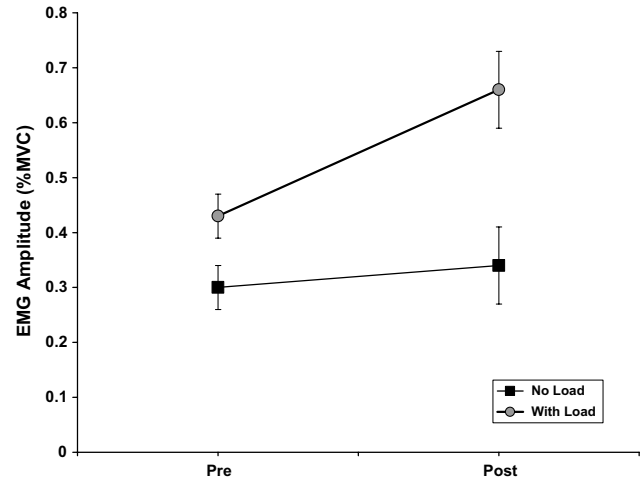

Fig. 4. Interaction of load and time on lumbar EMG amplitudes. Repetition caused an increase only in the loaded condition. The mean and standard error are shown ($n = 21$).

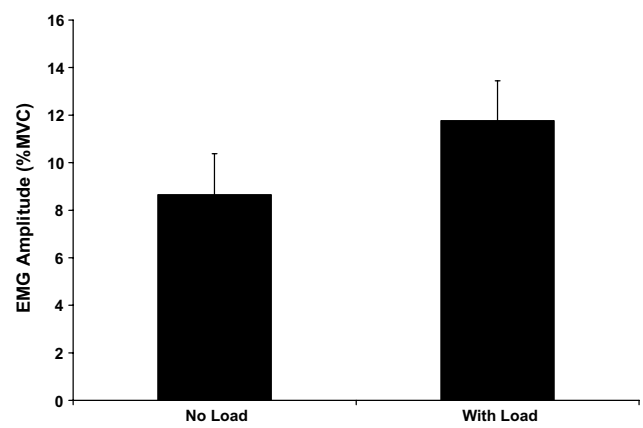

Fig. 5. Thoracic erector spinae activation as a function of load showing the statistically significant difference due to load. There were no significant gender or time effects. The mean and standard error are shown ($n = 42$).

For both the LES and TES EMG amplitudes, there was no difference between the left and right channels so these data were pooled. In terms of the muscular activation levels during flexion–relaxation, by definition the lumbar sites were consistently activated less than one percent MVC. Despite the low activation levels, there were statistically significant effects of load ($P < 0.001$), timing ($P < 0.01$) and a significant load $*$ timing interaction ($P < 0.05$). There was no effect of repetition on LES activity in the unloaded condition, but repetitive loading resulted in an increase in LES EMG amplitude when the load was held in the hands (Fig. 4). The thoracic sites showed no significant effects of repetition or gender (justifying collapse across these variables), and a significant increases in activation with load ($P < 0.05$; Fig. 5).

4. Discussion

We observed significant changes in the flexion–relaxation response with repeated spinal flexion. Both the maximum flexion angle and the flexion–relaxation angle parameters showed increases due to repeated spinal flexion, although the change in maximum flexion was not significant. There was also a significant interaction between time and load as the changes due to repeated spinal flexion were larger for the no load compared to the loaded condition. We observed a significant increase in the flexion–relaxation angle expressed as a proportion of full flexion. Repeated flexions also were associated with increases in the level of thoracic erector spinae activation and increases in lumbar erector spinae activation in the loaded conditions. Although the increases in flexion

angle are modest, they are thought to be meaningful due to the steep moment–angle relationship near full flexion and resulting significant changes in moment accompanying modest changes in angle.

Myoelectric deactivation near full flexion (flexion–relaxation) is thought to occur due to the passive tissues being stretched to the point that they can support the external spinal moment (Schultz et al., 1985; Kippers and Parker, 1984; Toussaint et al., 1995). This theory suggests that the flexion–relaxation angle should reflect the external moment, and supportive evidence has been presented for comparisons of sitting versus standing (Callaghan and Dunk, 2002). Most reports studying the effects of loads on flexion–relaxation show increases in spinal flexion with load (Schultz et al., 1985; Kippers and Parker, 1984; Gupta, 2001; Shirado et al., 1995), though few reports show no change (Sarti et al., 2001). We observed that the amount of load modulates the maximum flexion and flexion–relaxation angles due to repeated flexion; the unloaded condition showed greater changes due to repeated flexion than the load condition. In fact, the maximum flexion angle for the loaded condition was similar before and after repeated flexion; 46.5° (SE 2.1) versus 46.6° (SE 2.0), respectively, while there was an increase in the unloaded condition after repeated flexion; 44.5° (SE 1.9) versus 46.6° (SE 1.9), respectively. The changes in unloaded maximum angle reflect the expected increase in spinal creep due to repeated loading (Adams and Dolan, 1996). We expected a similar increase in loaded condition (due to creep), however, we did not observe any change in maximum flexion angle for the loaded condition before and after repeated flexion. Similarly, in terms of the flexion–relaxation angle, the increased angle due to repeated flexion was greater in the unloaded condition, contrary to the expected changes due to the passive tissue stress–strain relationship. The expected changes in the passive tissue mechanical properties due to repeated flexion do not fully explain the observed trends. It would seem that the repeated flexion resulted in concurrent changes in the stress–strain relationship of the passive tissues and the neuromuscular control system.

Researchers report increases in flexion–relaxation angle between 1.5° (Gupta, 2001) and 3° (Kippers and Parker, 1984) when subjects held a 10 kg mass in their hands, while we observed similar increases at the start of the experimental session (1.9°) and no difference in flexion–relaxation angle after the repeated trunk flexion. We observed that flexion–relaxation occurred at 92.7% of maximum flexion and was unaffected by load. This was similar to Sarti et al. (2001) who found similar loaded and unloaded values of 90–91% for fast loading (5 s flexion cycle) but smaller values of 84–86% for slow loading (7 s flexion cycle). It is possible that during fast loading, the reflex time allows for larger flexion angles before flexion–relaxation occurs. It was interesting to note that flexion–relaxation occurred closer to maximum flexion (95.2%) after repeated flexion.

We observed that the response of the maximum flexion angle to the loads did not depend on gender. Few studies have examined the effect of gender on spinal motions; many studies only evaluate the flexion–relaxation response of either male (Portnoy and Morin, 1956; Tanii and Masuda, 1985; Morris et al., 1962) or female subjects (Leinonen et al., 2000), while other studies evaluate both male and female subjects but do not test for gender effects (Gupta, 2001; Sihvonen et al., 1991; Ahern et al., 1988; Shirado et al., 1995). Callaghan and Dunk (2002) observed no significant differences in the flexion range of motion and the flexion–relaxation angle between genders, similarly to our study. However, they did observe that the females had a larger difference in the angle of myoelectric resumption between sitting and standing (Callaghan and Dunk, 2002). In addition, studies have noted that males have significantly greater spinal flexion range of motion (Wolf et al., 1979; Marras et al., 1994), velocity and acceleration (Marras et al., 1994). Marras et al. (2002) recently reported that women had greater relative spinal loads due to differences in body mass and muscle cocontractions, though we did not observe significant changes in flexion–relaxation angle nor flexion–relaxation angle as a percentage of maximum between the male and female subjects.

The flexion–relaxation angle did not differ between the mass and no-mass conditions after the repetitive flexion tasks. This observation is not consistent with the hypothesis that flexion–relaxation is due to tension in the passive tissues which would predict a significantly larger flexion–relaxation angle in the mass condition. Recent neurophysiology studies have demonstrated complex interaction between the dorsal spinal ligaments (supraspinous ligaments and the facet joint capsules) and the paraspinal muscles (Holm et al., 2002). Elizabeth Eversull et al. (2001) have demonstrated that the reflex activation of the multifidus muscle is dependent on the rate and magnitude of supraspinous ligament stretch and is modified by cyclic loading (Solomonow et al., 1999) and load rate (Solomonow et al., 2001). The facet joint afferents influence the excitability of the paraspinal musculature in animal models (Indahl et al., 1997; Kang et al., 2002). Although our observations from the beginning of the experiment indicated that increased external load was associated with an increased flexion–relaxation angle, this relationship was changed after repetitive trunk flexion. It appears that the flexion–relaxation phenomenon is not solely a function of passive tissue loading. The results of the current experiment are not entirely consistent with either a stress nor strain-dependent hypothesis for onset of flexion–relaxation. It is likely that there is a dependence upon neuromuscular reflexes, and that these reflex responses may be modified as a result of repetitive trunk flexion movements.

While the superficial erector spinae show flexion–relaxation, Andersson et al. (1996) report that the quadratus lumborum and deep erector spinae become more active with full trunk flexion. It is possible that the lumbar erector spinae muscle activity that we report in the current study may be due to activity of these deep muscles.

These results are based on a relatively large number of subjects, however they were all young and healthy. There are significant changes in spinal range of motion with age, and an interaction between age and gender (Marras et al., 1994); accordingly, these results may not be directly applicable to other age groups. The current experiment involved 100 repetitions of an unconstrained task from standing to full flexion. This task does not represent realistic industrial tasks. Only one task rate was evaluated in the current experiment; Sarti et al. (2001) report an increase in flexion–relaxation angle as the lift rate increases, and Mathieu and Fortin (2000) report a greater occurrence of flexion–relaxation with slower flexion rates.

There is considerable support for the idea that the onset of flexion–relaxation occurs at the point when the spinal passive tissue stretch balances the external spinal moment (Schultz et al., 1985; McGill and Kippers, 1994; Toussaint et al., 1995). Several studies document that changes in posture (Callaghan and Dunk, 2002; Gupta, 2001) and added external mass leads to expected increases in the flexion–relaxation angle (Schultz et al., 1985; Kippers and Parker, 1984; Gupta, 2001), though others present evidence of no change due to added mass (Sarti et al., 2001). We observed that repetitive trunk flexion modulates the flexion–relaxation angle with added external mass. Accordingly, it would appear that some factor other than simple spinal tissue stretch is involved; it is likely that changes to the neuromuscular reflex system are occurring, perhaps with the deeper, unmonitored muscles.

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